PF233-CB1D: CB1D-B controller: Pico 2 W, supplies, handle cable, receive-board links, TX monitor

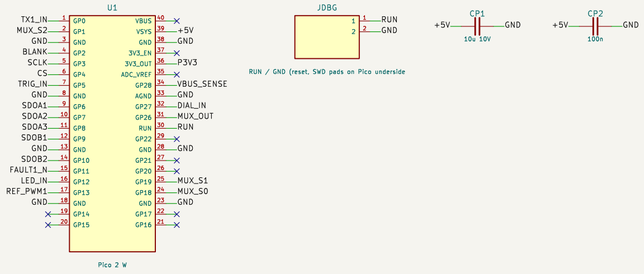
How to read: parts sit in functional blocks (dashed frames); every pin ends in a short wire with its NET NAME; equal names are connected (also between blocks). A cross = intentionally unconnected pin.

Diode symbols point anode to cathode. Pin names are those of the library; items marked VERIFY in docs/ASSUMPTIONS.md are unverified. MODELED design: not built, not bench-tested.

1 Controller

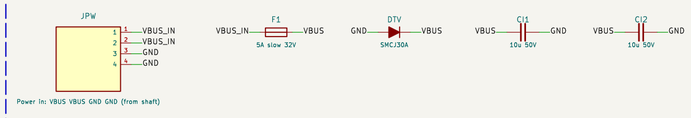
U1: Raspberry Pi Pico 2 W (RP2350 + CYW43439). Powered from +5V at VSYS; 3V3_OUT = P3V3.

JDBG: RUN / GND reset pads (SWD pads are on the Pico underside). CP1 / CP2 decouple VSYS.



2 Power input and protection

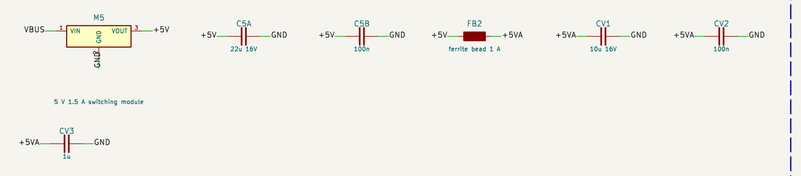
JPW: VBUS x2 / GND x2 from the shaft -> F1 5 A slow-blow -> VBUS (DTV: 30 V TVS, CI1 / CI2 10 uF bulk).



3 5 V and analog supplies

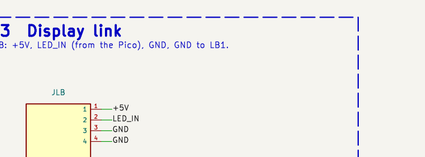
M5 (OKI-78SR-5 class, datasheet unverified) makes +5V from VBUS (C5A / C5B).

FB2 ferrite bead separates +5VA (CV1 / CV2 / CV3) for the receive boards.



4 3.3 V for the ADC I/O

UL (AMS1117-3.3 type) makes +3V3D from +5V for the digital side of the ADCs on AB1A / AB1B.



5 Monitor multiplexer

UM (74HC4051): S0 / S1 / S2 from the Pico, MUX_COM -> RMO / CMO -> the Pico ADC (MUX_OUT).

Inputs in use: Y0 = ISENSE1, Y1 = NTC_SENSE, Y4 = HTA_IN; E, VEE and GND are grounded.

6 Current-limit reference (REF_I1)

REF_PWM1 (PWM from the Pico) -> RR1 -> REF_N1 -> RR2 -> REF_I1 with CR1 / CR2 to GND: a two-pole RC filter; REF_I1 goes to the comparator on PB1R.

7 VBUS sense

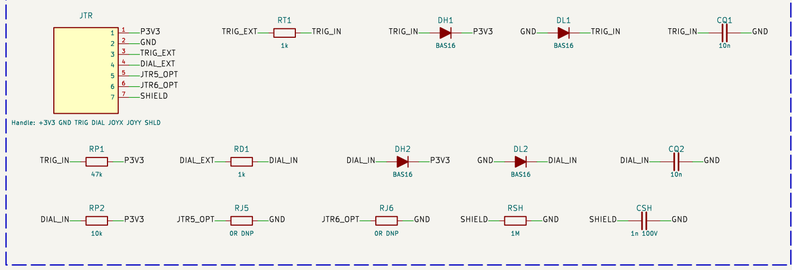
RV1 100 k / RV2 10 k divider from VBUS to VBUS_SENSE (Pico ADC), CV4 filter.

8 Handle cable: trigger, dial, shield

JTR: +3V3, GND, TRIG, DIAL, JOYX, JOYY, SHLD (the only signal cable leaving the head).

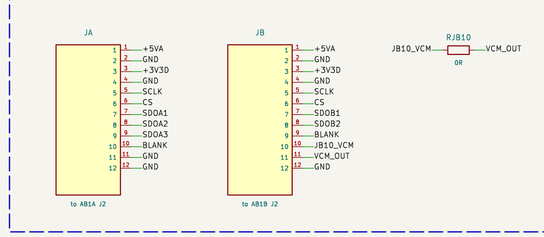
TRIG / DIAL inputs: series 1 k, BAS16 clamps to P3V3 / GND (DH / DL), 10 nF, pull-ups RP1 47 k / RP2 10 k.

RJ5 / RJ6 (0 R, not fitted): optional grounding of JTR.5 / JTR.6. RSH + CSH shield termination.



9 Links to the receive boards AB1A / AB1B

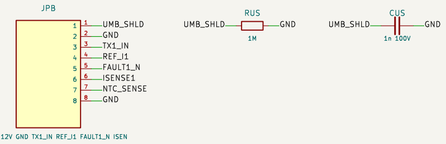
JA / JB: +5VA, +3V3D, SPI (SCLK, CS, SDOA1..3, SDOB1..2), BLANK, and from AB1B the VCM_OUT return; RJB10 (fitted 0 R) ties JB.10 to VCM_OUT.



10 Link to PB1R

JPB: 12V-side signals only: UMB_SHLD, TX1_IN, REF_I1, FAULT1_N, ISENSE1, NTC_SENSE, GND (same order as JC on PB1R).

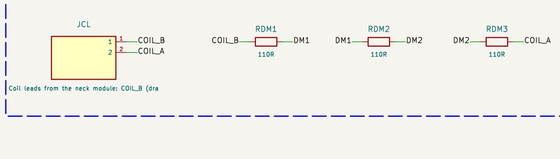
RUS / CUS bleed the umbilical shield.



11 Coil monitor: damping chain

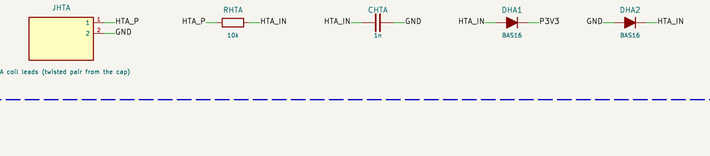
JCL: coil leads COIL_B (drain side) / COIL_A (VBUS side). RDM1..3 (3 x 110 R, 600 V) form the damping chain DM1 / DM2 between them.

High-voltage nets COIL_B / DM1 / DM2: 1.0 mm rule.



12 HTA timing-coil input

JHTA: HTA coil leads (twisted pair from the cap). RHTA 10 k series, CHTA 1 n, DHA1 / DHA2 clamp to P3V3 / GND -> HTA_IN (multiplexer input).



13 Display link

JLB: +5V, LED_IN (from the Pico), GND, GND to LB1.

Generated: src/pcb/sch_func.py; checked against the netlist.
MODELED design: not built, not bench-tested.
CB1D-B controller: Pico 2 W, supplies, handle cable, receive-board links, TX monitor
ARX / PHI-FOLD 233

Sheet: /	
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Title: PF233-CB1D functional schematic	
Size: User	Date: 2026-10-26
KiCad E.D.A. 9.0.9	**Rev: r3**
	Id: 1/1